Contents

1. What's the Deal with Matrices?

So, datastructures. By now, hopefully we're familiar with scalars and vectors. Scalars themselves are a 0d point, and they can describe 1d space. The number line. Vectors themselves are 1d lists of scalars, and they can describe 2d+ space. Graphs.

But vectors are datastructures which contain other datastructures. It's a storage type which stores scalars. What would happen if we had a vector containing other vectors, though? (Strictly speaking, we aren't allowed to do this. Imagine for a moment, though, that we could.)

$$\vec{v} = a\hat{\imath} + b\hat{\jmath} = [a, b]$$
$$\vec{u} = [c, d]$$
$$\vec{w} = [\vec{v}, \vec{u}] = [[a, b], [c, d]] \approx \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ or } \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Essentially, what we would have is some 2d collection of scalars. An *area* of scalars instead of a *line*. Rather than an nd vector, we have something like an $(m \times n)$d vector.

Well, we don't call it a vector. Instead, we call it a matrix.

A `(term)` matrix is just another type of datastructure, to borrow terminology from computer science. We can represent information using scalars, we can represent it using vectors, and we can represent it using matrices. Consequently, each of those tools have many associated applications and techniques. Here, we're going to learn about the ones associated with matrices.

(Aside: To extend the analogy further, we could also have matrices of matrices and so forth. 3d+ storage structures. But that's beyond the scope of this document.)

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`(TODO)` important to think of as separate from vectors too

Now, just as we can represent a scalar using a 1d vector, we can represent vectors as $1 \times n$ or $n \times 1$ matrices.

$$\vec{v} = a\hat{\imath} + b\hat{\jmath} + c\hat{k} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} a & b & c \end{bmatrix}$$

`(TODO)` col & row vectors

`(TODO)` describe 1d, describe 2d and beyond, describe what? Describe system of scalars, describe system of vectors. Cross product is a matrix thing.

2. Binary Operations

Scalar-matrix multiplication and matrix addition/subtraction are trivial.

$$k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}$$

Notice that matrix addition is only defined between equally-sized matrices.

Matrix-scalar addition is formally undefined, although it is sometimes useful to allow:

$$k + \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} k+a & k+b \\ k+c & k+d \end{bmatrix}$$

Matrix multiplication is more complicated and explained below.

Matrix division is treated as an extension of matrix multiplication, in the same way that scalar division is multiplication by a negative exponent. It will be discussed separately as the "Inverse".

2.1. Matrix Multiplication

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} \neq \begin{bmatrix} ae & bf \\ cg & dh \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$$

$$\begin{bmatrix} a & b \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac+bd \end{bmatrix}$$

`(technique)` The procedure to do matrix multiplication is as follows:

0. Given $\boldsymbol{AB} = \boldsymbol{C}$, find $\boldsymbol{C}$.
1. Know that $\boldsymbol{C}$ will have the height of $\boldsymbol{A}$ and the width or $\boldsymbol{B}$.
2. The $(i \times j)$th element of $\boldsymbol{C}$ will be the summation of the piecewise multiplication of the ith row of $\boldsymbol{A}$ with the jth column of $\boldsymbol{B}$.
3. Repeat the prior step for each element in $\boldsymbol{C}$.

From this we can see that matrix multiplication is defined if and only if the width of $\boldsymbol{A}$ equals the width of $\boldsymbol{B}$.

Further, recognize that matrix multiplication is not commutative.

$$\boldsymbol{AB} \neq \boldsymbol{BA}$$

2.1.1. The Identity Matrix

$$[1], \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \dots$$

The `(term)` identity matrix is a square matrix with 1s along its diagonal and the rest zero, as shown above. It is usually denoted with $\boldsymbol{I}$, or sometimes $\boldsymbol{I}_n$ where it is an $n \times n$ matrix.

Consider that, with scalar-scalar multiplication, we have the concept of "1", where anything times 1 is equal to iteslf. The identity matrix is matrix-multiplication's equivalent. (You can do the matrix multiplication to show this, if you'd like.)

$$\boldsymbol{A}\boldsymbol{I} = \boldsymbol{A}$$
$$\boldsymbol{I}\boldsymbol{A} = \boldsymbol{A}$$

3. Unary Operations

3.1. The Determinant

$$\text{let } \boldsymbol{A} \text{ be any } \underline{\text{square}} \text{ matrix}$$

$$\text{determinant of } \boldsymbol{A} = \det(\boldsymbol{A}) = |\boldsymbol{A}| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

The `(term)` determinant is to a matrix as the magnitude is to a vector. It is a scalar which represents, in some sense, the "size" of the matrix. Consider:

$$|C\boldsymbol{A}| = C|\boldsymbol{A}|$$
$$|\boldsymbol{A}\boldsymbol{B}| = |\boldsymbol{A}|\ |\boldsymbol{B}|$$

Where the magnitude of a vector is its length, or its 1d size, the determinant of an $n \times n$ matrix is the nd hypervolume whose boundaries are described by the matrix's row or column vectors. Good luck visualizing that.

Consider, consequently, why the matrix must be square. In order to have enough dimensionality for each row/column to create boundaries for an nd space, they must have n components.

The major difference, though, is that the determinant may be positive or negative. (`(TODO)` meaning of that)

`(TODO)` other implications, collinearity

Unfortunately, computing determinants is harder than magnitudes. We define the process recursively. First, the "base cases" are the trivial determinants of 1×1 and 2×2 matrices:

$$|[a]| = a$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$$

To remember the 2×2 matrix, consider that we just multiplied across the diagonals and subtracted the terms.

I will present the 3×3 and 4×4 determinants before explaining the process below.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$\begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix} = a\begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b\begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c\begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d\begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

`(technique)` Now, to compute the determinant:

0. We are given some $n \times n$ matrix and asked to find its determinant, where $n \geq 3$.
1. Pick some row or column as our top-level scalar multipliers. (In the examples above, the first row (abcd) was chosen both times. We could have just as easily picked efgh or cgko.) We will iterate through that row/column.
2. Each item in that row/column corresponds to a term in the output. The term consists of that item multiplied by the determinant of the $(n-1) \times (n-1)$ matrix formed by omitting that item's row and column.
3. Select the correct sign for the term according to the checkerboard $\pm$ pattern explained below.
4. Repeat for each item in the initially selected row/column.
5. We now have a problem involving $(n-1) \times (n-1)$ matrices. Repeat until we have 2×2 matrices which we can resolve with the known formula.

`(TODO)` better visuals

The checkerboard pattern is demonstrated:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \qquad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix} \qquad \begin{bmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{bmatrix}$$

`(technique)` Consider that, since we can select any row/column as our starting point, it is advantageous to select that row/column which contains the most zeroes, since we know a zero's corresponding term will zero-out.

3.2. The Transpose `(TODO)`

$$\text{let } \boldsymbol{A} \text{ be any matrix}$$

$$\text{transpose of } \boldsymbol{A} = \boldsymbol{A}^T$$

$$\text{if } \boldsymbol{B} = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \text{ then } \boldsymbol{B}^T = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}$$

$$\left(\boldsymbol{A}^T\right)^T = \boldsymbol{A}$$

`(TODO)` symmetric, skew symmetric

3.3. The Adjugate (or Classical Adjoint) `(TODO)`

$$\text{let } \boldsymbol{A} \text{ be any } \underline{\text{square}} \text{ matrix}$$

$$\text{adjugate of } \boldsymbol{A} = \operatorname{adj}(\boldsymbol{A})$$

The terms `(term)` adjugate and `(term)` classical adjoint are synonyms.

3.4. The Trace **(TODO)**

3.5. The Inverse **(TODO)**

4. Row Operations **(TODO)**

5. Eigenvalues and Eigenvectors **(TODO)**

6. Diagonalization and Exponentiation **(TODO)**

7. Laplace Tranaformations **(TODO)**